PTI STUDY GUIDE - MODULE 0: TREE BIOLOGY FUNDAMENTALS

Introduction

Before measuring defects, using assessment tools, or making management decisions, you must understand **what you're assessing**. Trees are complex living organisms with sophisticated structural and physiological systems. This module covers the fundamental biology that underlies all professional tree assessment.

Understanding tree anatomy and physiology allows you to:

- Interpret why defects develop
- Predict how defects will progress
- Understand species-specific responses
- Recognize adaptive vs maladaptive changes
- Make evidence-based management decisions

This is the foundation for all subsequent modules.

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1. EXTERNAL STRUCTURE

The Three Major Components

Every tree consists of three integrated systems:

Crown (aerial portion):

- Branches and foliage
- Primary function: Photosynthesis (energy production)
- Secondary functions: Reproduction, hormone production
- Structural role: Creates loading on stem and roots

Stem/Trunk:

- Main structural axis
- Primary function: Support (elevates crown)
- Secondary functions: Storage, transport, defense
- Connects crown to roots (structural and vascular)

Root System:

- Below-ground portion
- Primary function: Anchorage (prevents windthrow)
- Secondary functions: Water/nutrient uptake, storage
- Typically extends 2-3× crown radius

Crown Architecture

Apical Dominance:

- Terminal bud produces hormones (auxins) that suppress lateral buds
- Creates single dominant leader in many species (excurrent form)
- When apical dominance lost → multiple co-dominant stems (decurrent form)

Branch Angles:

- Angle affects attachment strength
- Wide angles (60-90°): Stronger attachment, bark ridge visible
- Narrow angles (<45°): Weaker, risk of included bark
- Species-specific: Some naturally have narrow angles (poplars), others wide (oaks)

Crown Distribution:

- Even distribution = balanced loading
- Asymmetric crown = eccentric loading (tree compensates with reaction wood)
- Crown density = indicator of tree vigor

Stem Form

Taper:

- Gradual diameter decrease from base to apex
- Optimal taper: Stress evenly distributed
- Uniform taper = healthy structural development
- Abrupt taper change = possible internal defect

Buttressing:

- Swelling at base where stem meets roots
- Increases leverage resistance at ground line

- Well-developed buttresses = good structural adaptation
- Species variation: Some buttress heavily (London plane), others minimally

Reaction Wood Formation:

- Trees add material on stress side
- Compression wood (conifers): On compression side of lean
- Tension wood (broadleaves): On tension side of lean
- Creates eccentric growth rings (wider on stress side)

Root Architecture

Structural Roots (woody, large diameter):

- Provide anchorage
- Typically radiate from stem (4-8 major roots common)
- First 1-3m critical for stability (root plate)
- Depth: Usually shallow (top 1m), species and soil dependent

Transport Roots (medium diameter):

- Extension of structural roots
- Transport water and nutrients
- Branch repeatedly, forming network

Absorbing Roots (fine roots, <2mm):

- Ephemeral (short-lived, constantly replaced)
- Have root hairs for absorption
- Located in top 30cm of soil typically
- Form mycorrhizal associations (fungal partnerships)

Root Collar:

- Transition zone where stem becomes roots
- Often flared (buttressed)
- Critical inspection area (decay entry point, girdling roots)
- Should be visible at soil surface (not buried)

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2. INTERNAL ANATOMY

The Layered Structure

Trees grow in **concentric layers** from inside out. Understanding this layering is essential for interpreting decay, wounds, and structural defects.

Bark (Outer Protection)

Outer Bark (Rhytidome):

- Dead tissue
- Protection from: Fire, mechanical damage, pests, UV radiation, temperature extremes
- Sheds as tree expands (species-specific patterns: flaky, furrowed, plated)
- No living cells, no regeneration

Inner Bark (Phloem):

- Living tissue (thin layer, usually 1-5mm)
- Function: Transports sugars from leaves to rest of tree
- Direction: Downward and lateral (from crown to roots and storage tissues)
- Vulnerable: Damage here (girdling) can kill tree above damage point
- Replaced seasonally from cambium

Cork Cambium (Phellogen):

- Meristematic layer (produces new cells)
- Creates outer bark (outward) and phelloderm (inward)
- Allows bark to expand as tree grows
- Activated when inner bark or cambium damaged (wound periderm)

Cambium (Growth Layer)

Vascular Cambium:

- Single-cell-thick layer
- Location: Between bark (phloem) and wood (xylem)
- Function: Produces ALL secondary growth (trunk diameter increase)
- Division pattern:
    - Divides outward → new phloem cells (inner bark)

- Divides inward → new xylem cells (wood)
- Ratio: ~10:1 (produces much more wood than bark)

Growth Rhythm:

- Active: Spring through autumn (temperate zones)
- Dormant: Winter
- Creates annual rings (one ring = one year's growth)

Cambium Continuity:

- Continuous sheet wrapping entire tree
- Connects crown to roots
- Critical for wound closure (callus growth originates from cambium)
- Damage to cambium = permanent scar (cambium cannot regenerate across dead tissue)

Xylem (Wood)

Wood is composed of **xylem tissue** - dead cells forming structural matrix and transport system.

Sapwood (Active Xylem):

- Outer wood (adjacent to cambium)
- Function: Water transport from roots to leaves
- Living cells present (parenchyma cells for storage/defense)
- Light colored (moisture content 50-200% of dry weight)
- Width varies: 2-5cm typical in temperate trees, up to 15cm in fast-growing species

Heartwood (Inactive Xylem):

- Inner wood (toward center)
- Function: Structural support only (no transport)
- No living cells (all cells dead)
- Darker colored (from extractives: tannins, resins, oils)
- More durable (extractives resist decay)
- Width increases with age (sapwood converts to heartwood progressively)

Transition:

- Sapwood → Heartwood conversion occurs gradually
- Tree withdraws nutrients, deposits extractives
- Process controlled by tree (not passive aging)
- Rate varies by species: Fast (ash, willow) vs slow (beech, holly)

Structural Properties:

Both sapwood and heartwood contribute to strength (dead cells still structural), but:

- **Sapwood:** Can decay rapidly (moisture + nutrients support fungi)
- **Heartwood:** More decay-resistant (extractives inhibit fungi)
- **Implication:** Sapwood decay often more aggressive than heartwood decay

Annual Rings

Ring Formation:

Each year's growth appears as distinct ring (visible in cross-section):

- **Earlywood (Springwood):** Formed early in growing season
  - Large cells, thin walls
  - Function: Maximize water transport (spring demand)
  - Light colored, less dense
- **Latewood (Summerwood):** Formed late in growing season
  - Small cells, thick walls
  - Function: Structural strength
  - Dark colored, more dense

Ring Pattern: Earlywood + Latewood = One annual ring

Reading Rings:

Width indicates growth rate:

- **Wide rings:** Favorable conditions (adequate water, nutrients, light)
- **Narrow rings:** Stress (drought, defoliation, competition, root damage)
- **Sudden change:** Environmental event (e.g., adjacent tree removed → release)

Species Differences:

Ring-porous (oak, ash, elm, chestnut):

- Large vessels concentrated in earlywood (distinct ring boundary)
- Latewood vessels much smaller
- Current year's earlywood = primary water transport
- **Implication:** Girdling damage severe (cannot transport water if current year's ring damaged)

Diffuse-porous (beech, birch, maple, sycamore):

- Vessels uniform size throughout ring
- Multiple years' rings functional for transport
- More redundancy in vascular system
- **Implication:** Girdling damage less immediately severe (other rings compensate)

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3. VASCULAR SYSTEM

Trees have two separate transport systems moving materials in opposite directions.

Xylem Transport (Water & Minerals)

What moves: Water + dissolved minerals (nitrogen, phosphorus, potassium, etc.)

Direction: Upward (roots → leaves)

Mechanism: Transpiration-Cohesion-Tension Theory

1. **Transpiration:** Water evaporates from leaves (stomata)
2. **Tension:** Creates negative pressure (suction) in leaf xylem
3. **Cohesion:** Water molecules stick together (hydrogen bonds)
4. **Column:** Continuous water column pulled upward
5. **Root uptake:** Water enters roots to replace what's lost

Driving force: Solar energy (powers transpiration)

Flow rate:

- Can reach 20-30 meters/hour
- Depends on: Tree size, leaf area, temperature, humidity, soil moisture

Vulnerability:

- **Cavitation:** Air bubbles break water column (freeze-thaw cycles, drought stress)

- **Embolism:** Blocked vessel (air pocket prevents flow)
- **Tree response:** Redundant vessels, can seal embolized vessels

Phloem Transport (Sugars)

What moves: Sugars (primarily sucrose) + hormones + signaling molecules

Direction: Source to sink (typically downward, leaves → roots, but can be lateral or upward)

Mechanism: Pressure Flow (Mass Flow)

1. **Loading:** Sugars loaded into phloem at source (leaves)
2. **Pressure:** High sugar concentration draws water in (osmosis)
3. **Flow:** Positive pressure pushes solution through phloem
4. **Unloading:** Sugars unloaded at sink (roots, storage tissues, growing points)

Sinks (where sugars go):

- Growing tissues (buds, cambium, root tips)
- Storage tissues (ray cells, root reserves)
- Repair tissues (wound closure, reaction wood)
- Reproductive structures (flowers, seeds, fruit)

Vulnerability:

- **Girdling:** Physical damage around circumference removes phloem
- **Effect:** Sugars cannot reach roots → root starvation → reduced water uptake → tree decline
- **Timeline:** Symptoms may take months to years (depends on stored reserves)

Integration of Systems

Xylem and phloem work together:

- Xylem supplies water for photosynthesis
- Photosynthesis produces sugars transported in phloem
- Sugars support cambium growth → more xylem and phloem
- Feedback loop: Each system depends on the other

When vascular systems disrupted:

- Xylem damage: Wilting, dieback, drought stress symptoms

- Phloem damage: Reduced growth, root decline, eventual death
- Both damaged: Rapid decline (combined stress)

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4. COMPARTMENTALIZATION OF DECAY IN TREES (CODIT)

The CODIT Concept

Developed by Dr. Alex Shigo (1970s-1980s), CODIT explains how trees **resist the spread of decay** after wounding.

Key principle: Trees cannot heal (regenerate damaged tissue). Instead, they **compartmentalize** (wall off) damage.

Misconception: "Trees heal wounds by growing over them."
Reality: Trees seal wounds with new growth, but damaged tissue remains (compartmentalized, not healed).

The Four Walls of CODIT

Trees create chemical and physical barriers in four directions:

Wall 1: Resistance vertically (up/down the stem)

- Weakest wall
- Formed by plugging vessels with tyloses and gums
- Decay spreads easily up and down (within annual ring)
- **Result:** Decay column extends above and below wound

Wall 2: Resistance radially inward (toward pith)

- Moderate strength
- Formed by growth ring boundary (latewood dense)
- Each annual ring acts as partial barrier
- Decay advances slowly inward (year by year, ring by ring)
- **Result:** Decay progresses toward center over time

Wall 3: Resistance tangentially (around circumference)

- Moderate strength
- Formed by ray cells (radial wood strands)
- Decay spreads around circumference within single ring
- **Result:** Decay expands around stem within same ring

Wall 4: Resistance radially outward (new growth)

- Strongest wall
- Formed by new wood (cambium produces barrier zone)
- New growth laid over wound = physical separation
- **Critical:** Wound occurs in year X; growth in year X+1 = Wall 4
- **Result:** Decay rarely penetrates beyond wound year (if Wall 4 intact)

Wall 4 - The Critical Barrier

Barrier Zone Formation:

When cambium detects wound:

1. Produces specialized cells in next year's growth
2. Barrier zone cells: Smaller, denser, more extractives
3. Creates impermeable layer between old wood (wounded) and new wood (healthy)

Callus Growth:

- New tissue grows from wound margins (from cambium)
- Covers wound surface progressively (years to complete in large wounds)
- Does NOT regenerate destroyed tissue
- Functions: Protects exposed wood, reestablishes bark continuity

CODIT Success Depends On:

Tree vigor:

- Healthy tree: Strong barrier zone, fast callus growth
- Stressed tree: Weak barriers, slow closure, decay advances

Wound size:

- Small wounds: Close quickly, minimal decay
- Large wounds: Slow closure, decay has time to establish

Wound timing:

- Growing season: Cambium active, immediate response
- Dormant season: Delayed response (wait until spring)

Pathogen type:

- Opportunistic fungi: Contained by CODIT (most wood decay fungi)
- Aggressive pathogens: Breach barriers (e.g., Kretzschmaria deusta)

Species variation:

- Strong compartmentalizers: Oak, sweetgum, maple
- Weak compartmentalizers: Poplar, willow, elm

CODIT in Practice

Pruning implications:

- Flush cuts: Remove branch collar (destroys natural barrier) → decay enters main stem
- Proper cuts: Leave collar intact → decay contained in branch stub
- Wound dressing: NOT recommended (doesn't improve CODIT, may trap moisture)

Cavity development:

- Decay contained by CODIT walls initially
- Over time (years/decades), decay column expands
- Hollow forms as decay advances, outer shell remains (if Wall 4 intact)
- **Result:** Tree with internal cavity but functional exterior shell

Management considerations:

- Cannot reverse decay (CODIT contains, doesn't eliminate)
- Avoid creating unnecessary wounds (each = potential decay entry)
- Support tree vigor (improves CODIT effectiveness)
- Assess shell thickness, not just presence of decay

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5. WOUND RESPONSE AND HEALING

The Wounding Process

Immediate effects:

- Cambium destroyed in wounded area

- Sapwood exposed
- Bark protection lost
- Vascular system disrupted

Short-term response (hours to days):

- Desiccation of exposed tissue
- Oxidation (surface darkens)
- Chemical defense activation (phenolic compounds, resins)

Medium-term response (weeks to months):

- Wound periderm formation (cork layer seals exposed tissue)
- CODIT barriers established
- Discoloration spreads in wounded wood

Long-term response (years):

- Callus growth from wound edges
- Gradual wound closure
- Barrier zone prevents decay spreading to new wood

Callus Formation

Process:

1. **Initiation:** Cambium at wound edge activated
2. **Woundwood production:** Irregular, curved growth pattern
3. **Expansion:** Callus grows over wound surface (from all edges simultaneously)
4. **Closure:** Edges meet (if wound not too large)
5. **Integration:** Callus smooths, bark reforms

Rate of closure:

Depends on:

- **Wound size:** Larger = slower (closure rate ~10-30mm/year radially)
- **Tree vigor:** Healthy = faster
- **Species:** Fast-growing species close wounds faster
- **Location:** Branches close faster than main stem

Examples:

- 50mm diameter wound: 2-5 years to close completely
- 200mm diameter wound: 10-20 years to close completely
- 500mm diameter wound: May never close completely

Partial closure:

- Large wounds often don't close fully
- Permanent opening remains
- Callus stabilizes wound (prevents expansion)
- Functional even if not cosmetically closed

Girdling

What is girdling?

Complete removal of bark (and cambium) around entire circumference of stem or branch.

Effects:

Immediate:

- Phloem transport severed
- Sugars cannot pass girdled section downward

Short-term (weeks-months):

- Roots receive no sugars
- Roots deplete stored reserves
- Root growth stops
- Water uptake reduces (fewer active roots)

Long-term (months-years):

- Crown shows stress (smaller leaves, reduced growth)
- Roots die back
- Tree becomes unstable (root anchorage weakens)
- Tree death (if girdle complete and not bridged)

Partial girdling:

- If 50-75% of cambium intact: Tree can survive
- Remaining cambium expands function
- May develop callus over girdled section eventually

Recovery potential:

- Complete girdle: Usually fatal (exceptions: species with extensive root grafts)
- Narrow girdle (<50mm): May bridge with callus growth
- Wide girdle (>100mm): Rarely bridges successfully

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6. REACTION WOOD AND ADAPTIVE GROWTH

How Trees Respond to Stress

Unlike animals with rigid skeletons, trees **actively modify their structure** in response to mechanical stress.

Reaction Wood Formation

When stem leans or is eccentrically loaded, tree adds extra wood on stress side:

Compression Wood (Conifers):

- Location: Underside of lean (compression side)
- Structure: Thick-walled cells, high lignin, more dense
- Color: Darker (reddish-brown)
- Function: Resists compression forces
- Growth pattern: Push upward (tries to straighten)

Tension Wood (Broadleaves):

- Location: Upper side of lean (tension side)
- Structure: Thick gelatinous layer (G-layer) in cell walls
- Color: Lighter (whitish)
- Function: Resists tension forces
- Growth pattern: Pull upward (tries to straighten)

Chronic Lean Indicators:

- Eccentric growth rings (wider one side)

- Reaction wood present on appropriate side
- **Interpretation:** Tree adapted to lean, likely stable
- **Contrast:** Recent lean without reaction wood = failure in progress

Thigmomorphogenesis

Definition: Growth response to mechanical stimulation (wind, movement, touch)

Effects:

- Trees in windy areas: Shorter, stockier, wider taper
- Trees in sheltered areas: Taller, thinner, less taper
- Swaying strengthens stem (within limits)

Practical implication:

- Trees suddenly exposed (adjacent tree removed) = maladapted to new wind regime
- May develop failure until adaptive growth compensates (takes years)

Adaptive Growth Patterns

Ribs and flutes:

- Vertical ridges along stem
- Tree adding material in compression zones
- Indicates appropriate response to loading

Overgrowth around defects:

- Swelling above/around cavity or crack
- Tree reinforcing weak point
- Indicates tree is coping (but defect still present)

Buttressing:

- Extra material at base
- Increases resistance to leverage (bending moment)
- Well-developed buttresses = sign of structural stability

Asymmetric crown reduction:

- Tree sheds branches on less-anchored side

- Reduces loading on weak roots
- Self-pruning as adaptive response

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7. SPECIES DIFFERENCES

Understanding species-specific characteristics is critical for accurate assessment.

Wood Density

Dense woods (strong, heavy):

- Examples: Oak, beech, ash, hornbeam
- Characteristics: Slow-growing, durable, decay-resistant
- Strength: High bending strength
- **Implication:** Can tolerate more decay (proportionally) than less dense species

Light woods (weak, lighter):

- Examples: Poplar, willow, horse chestnut
- Characteristics: Fast-growing, less durable, decay-prone
- Strength: Lower bending strength
- **Implication:** Decay more concerning (less structural reserve)

Decay Resistance

Durable species (heartwood):

- Examples: Oak, cedar, yew, sweet chestnut
- High extractive content (tannins, oils)
- Heartwood resists decay naturally
- Sapwood still vulnerable

Non-durable species:

- Examples: Beech, birch, lime, poplar
- Low extractive content
- Heartwood decays readily
- Rapid decay progression common

Compartmentalization Ability

Strong compartmentalizers:

- Examples: Oak, maple, sweetgum
- Form effective CODIT barriers
- Decay columns narrow, well-defined
- Wounds close relatively quickly

Weak compartmentalizers:

- Examples: Elm, poplar, willow
- Poor CODIT barriers
- Decay spreads extensively
- Wounds close slowly, decay progresses during closure

Branching Patterns

Excurrent (single leader):

- Examples: Most conifers, some broadleaves (sweet gum, tulip tree)
- Apical dominance maintained
- Strong central stem
- Branches clearly subordinate

Decurrent (multiple leaders):

- Examples: Oak, beech, elm
- Apical dominance lost early
- Crown divides into co-dominant stems
- Requires assessment of stem attachments

Root Characteristics

Deep-rooting species:

- Examples: Oak, pine, walnut
- Develop tap root or sinker roots
- Better drought tolerance
- More wind-firm (but species/site dependent)

Shallow-rooting species:

- Examples: Spruce, birch, beech, cherry
- Roots concentrated in top 50cm
- More susceptible to windthrow
- More sensitive to soil compaction

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8. INTEGRATION WITH TREE ASSESSMENT

How Biology Informs Assessment

Understanding anatomy → Interpreting defects:

- **Cambium damage → Girdling risk:** Assess circumference affected, predict timeline to decline
- **Sapwood decay → Transport disruption:** More urgent than heartwood decay in ring-porous species
- **Poor CODIT species → Decay progression:** Factor into monitoring intervals
- **Reaction wood → Chronic vs recent:** Distinguish stable adapted lean from progressive failure

Understanding physiology → Predicting outcomes:

- **Stressed tree + wound → Poor compartmentalization:** Decay likely to spread
- **Vigorous tree + pathogen → Effective response:** May contain infection
- **Root damage + drought → Compounded stress:** Higher failure risk
- **Girdling + time → Root decline → Instability:** Timeline depends on tree size and reserves

Questions Biology Helps Answer

"How serious is this decay?"

- Species decay resistance
- Heartwood vs sapwood
- CODIT effectiveness
- Time since infection

"How fast will this progress?"

- Species compartmentalization ability
- Tree vigor (growth rate, crown density)

- Pathogen aggressiveness
- Environmental stress

"Will this wound close?"

- Wound size relative to stem diameter
- Tree vigor
- Species growth rate
- Location (branch vs stem)

"Is this lean stable?"

- Reaction wood present? (Chronic = stable)
- No reaction wood? (Recent = unstable)
- Root architecture (deep vs shallow)
- Species flexibility

SUMMARY: KEY BIOLOGICAL PRINCIPLES

1. Trees Are Compartmented Organisms

- Vertical transport (xylem, phloem)
- Horizontal barriers (growth rings, rays)
- Decay contained, not eliminated
- Compartmentalization effectiveness varies

2. Structure Reflects Function

- Form follows mechanical requirement
- Abnormal form indicates abnormal loading/damage
- Adaptive growth = tree coping
- Lack of adaptation = concern

3. Trees Cannot Heal

- Dead tissue stays dead
- Wounds covered, not regenerated

- Decay remains (compartmentalized)
- Management = support tree's response

4. Species Matters

- Wood properties affect failure resistance
- Decay resistance varies widely
- Growth rate affects timeline
- Root architecture affects stability

5. Vigor Is Critical

- Vigorous trees compartmentalize better
- Stressed trees decline faster
- Multiple stresses compound
- Management supports vigor when possible

PRACTICE EXERCISES

Exercise 1: Anatomy Identification

Cross-section shows (from outside to inside): rough outer layer, thin brown layer, very thin line, light-colored wood (outer 3cm), dark-colored wood (center).

Question: Identify the five layers described.

Answer:

1. Rough outer layer = Outer bark (rhytidome)
2. Thin brown layer = Inner bark (phloem)
3. Very thin line = Vascular cambium
4. Light-colored outer wood = Sapwood (active xylem)
5. Dark-colored center = Heartwood (inactive xylem)

Exercise 2: CODIT Application

Branch pruned with flush cut 5 years ago. Wood inside pruning wound is decayed brown, extending 30cm into main stem. Wood outside original wound perimeter (grown since pruning) is sound.

Question: Which CODIT wall is functioning? Which wall was breached?

Answer:

- **Wall 4 functioning:** New wood (grown after wounding) is sound - barrier zone effective
- **Wall 1 breached:** Decay extended vertically 30cm (up/down) - weakest wall
- **Interpretation:** Wall 4 protecting new growth, but decay will continue in old wood (within Wall 4 boundary). Shell thickness determines structural safety.

Exercise 3: Species Comparison

Two trees, same size, same t/R ratio (0.28), same decay extent:

- Tree A: Oak (dense wood, strong compartmentalizer)
- Tree B: Poplar (light wood, weak compartmentalizer)

Question: Which presents higher risk? Why?

Answer: Tree B (poplar) presents higher risk because:

1. **Lower strength:** Light wood has less bending strength per unit volume
2. **Poor compartmentalization:** Decay likely progressing faster
3. **Less decay-resistant:** Both sapwood and heartwood decay readily
4. **Result:** Same t/R, but poplar has less residual capacity and worse prognosis

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Exercise 4: Wound Closure

Tree stem diameter: 400mm. Wound: 120mm wide. Tree is vigorous, adding 4mm diameter/year.

Question: Approximately how long until wound closes completely?

Answer:

- Wound radius: 60mm (120mm ÷ 2)
- Callus growth rate: ~2mm/year radially (from each side)
- Distance to close from each side: 60mm
- Time: 60mm ÷ 2mm/year = **30 years** (approximate)
- **Note:** This assumes sustained vigor and no complications - may be slower in reality

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NEXT MODULE

Module 1: Measurement Techniques

Now that you understand tree structure and function, Module 1 covers HOW to measure defects in that structure (cracks, decay extent, lean, crown condition).

This study guide provides the biological foundation for all professional tree assessment. Every defect you evaluate exists within the living system described here.